St. Francis Institute of Technology

(Engineering College)

An Autonomous Institute, Affiliated to University of Mumbai

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Department of Artificial Intelligence and Machine Learning

Academic Year: 2025-2026 **Term**: Even (Jan. 2026 – Jun. 2026)
Class / Branch: SE – AIML **Semester**: IV
Course: Web Programming Lab. (AI4VS_LR4)
Date of Assignment: / /2026 **Date of Submission:** / /2026

Pre-Lab Exercises for Experiment-9

Pre-Lab Activity 1: Understanding useEffect Execution

Task:

1. Create a functional component.
2. Display a simple message on screen.
3. Use `useEffect` to:
    - Print a message in the console when the component loads.

Pre-Lab Activity 2: Dependency Array Behavior

Task:

1. Create a counter using useState.
2. Add a useEffect that:
    - Logs "Counter Updated" whenever count changes.
3. Experiment with:
    - No dependency array
    - Empty dependency array
    - Dependency with count

Pre-Lab Activity 3: Simulating API Call using setTimeout

Before calling real APIs, simulate asynchronous behavior.

Task:

1. Create a component.
2. Create state called `data`.
3. Use useEffect to:
    - Simulate fetching data using setTimeout (2 seconds delay).
    - After delay, update state with sample data.
4. Show "Loading…" until data appears.

Pre-Lab Activity 4: Basic Fetch API (Without useEffect)

Task:

1. Create a button labeled “Fetch Data”.
2. On button click:
    - Fetch data from a public API.
    - Print response in console.

Pre-Lab Activity 5: Async/Await Practice

Task:

1. Create an async function.
2. Use async/await instead of .then().
3. Log fetched data in console.
4. Add try-catch for error handling.

Pre-Lab Activity 6: Rendering Array Data Dynamically

Task:

1. Create an array of objects manually.
2. Store it in state.
3. Use map() to display:
    - Name
    - Email
4. Render items in list format.

Pre-Lab Record Questions

1. When does useEffect execute?
2. What is the difference between .then() and async/await?
3. Why is loading state important during API calls?
4. Why must fetched data be stored in state?